Budget Analysis & Financial Planning

Veer Suraksha Grid

Smart Safety & Monitoring System for Sanitation Workers

Solapur Municipal Corporation (SMC)

2025–26

1. Introduction

This section presents a comprehensive budget analysis for the proposed Veer Suraksha Grid, a smart safety and monitoring system designed for sanitation workers under the Solapur Municipal Corporation (SMC).

The system integrates wearable safety monitoring and environmental sensing into a unified device to enhance worker safety, improve emergency response, and enable real-time governance.

The purpose of this analysis is to evaluate:

- Cost per worker device
- Scalability across deployment levels
- Operational and maintenance costs
- Cloud infrastructure requirements
- Financial feasibility within municipal budgets

The objective is to demonstrate that the system is cost-effective, scalable, and sustainable for large-scale public deployment.

2. System Cost Components

The total system cost is divided into the following categories:

- Hardware Cost (Per Device)
- Bulk Cost Optimization
- Optional Expansion
- Deployment Cost
- Operational Cost
- Software Infrastructure
- Total Cost of Ownership (TCO) — 5-Year Projection
- Deployment Timeline

3. Hardware Cost (Per Worker Device – Prototype)

The system combines VeerGuard (wearable safety) and VeerProbe (inspection capabilities) into a single integrated device, reducing redundancy and cost.

Component Cost Breakdown (Single Unit Prototype)

Sr.	Component	Qty	Unit Cost (₹)	Total Cost (₹)

1	ESP32 Microcontroller	1	400	400
2	MQ-2 Gas Sensor	1	120	120
3	MQ-4 Gas Sensor	1	100	100
4	MQ-7 Gas Sensor	1	120	120
5	DHT22 Sensor	1	100	100
6	Ultrasonic Sensor	1	70	70
7	MAX30102 Sensor (Heart Rate / SpO_2)	1	120	120
8	MPU6050 Sensor (Motion/Fall Detection)	1	120	120
9	Push Button (SOS Trigger)	1	20	20
10	Buzzer	1	20	20
11	LED Indicators	3	2	6
12	SX1276 LoRa Module	1	400	400
13	Wiring & Assembly	–	–	200
14	Battery & Power System	1	300	300
	Total Prototype Cost			**₹2,096 (~₹2,100)**

Prototype unit cost: approximately ₹2,096, rounded to ₹2,100 per device.

4. Hardware Cost Optimization Strategy

To ensure feasibility at scale, the following strategies are adopted:

- Bulk procurement directly from manufacturers
- Reduction of intermediary costs through vendor-level sourcing
- Integration of components into a single PCB system
- Optimization of communication modules
- Reduction in assembly and wiring complexity

These measures collectively enable substantial cost reduction during large-scale deployment.

4.1 Component-wise Bulk Cost Reduction Analysis

Bulk pricing estimates are based on supplier benchmarks (Robu.in, ElectronicsComp) and large-quantity procurement trends.

Component	Prototype (₹)	100 Units (₹)	1000 Units (₹)	Reduction (%)
ESP32 Microcontroller	400	300	220–250	40–45%
MQ Gas Sensors	100–120	75–90	60–70	40–45%
DHT22 Sensor	100	80	65	35%
Ultrasonic Sensor	70	55	45	35–40%
MAX30102 Sensor	120	90	70	40–45%
MPU6050 Sensor	120	90	70	40–45%
Push Button	20	15	10	50%
Buzzer	20	15	10	50%
LED Indicators	6	5	3	50%
SX1276 LoRa Module	400	250	180–200	50–55%
Integrated PCB System	200	120	80–100	50–60%
Battery System	300	240	180–220	30–40%

4.2 Key Cost Reduction Factors

LoRa Module Optimization:

At large procurement volumes, LoRa modules can be sourced directly from manufacturers or integrated at the chipset level. This results in a cost reduction of approximately 50–55%.

PCB System Integration:

Replacing discrete wiring with an integrated PCB reduces assembly complexity, improves durability, and lowers production cost by approximately 50–60%.

4.3 Final Optimized Cost

- Prototype Cost: ₹2,100 per device
- Bulk Cost (1,000 units): ₹950–₹1,100 per device

This represents an overall cost reduction of approximately 45–50%.

5. Optional Expansion Cost

For high-risk environments requiring advanced gas detection:

Component	Cost (₹)
MQ-136 Gas Sensor (H_2S Detection)	1,000

Updated Device Cost:

- Standard Device: ₹2,100
- Advanced Configuration: ₹3,100–₹3,200

This feature is deployed selectively in hazardous zones.

6. Deployment Cost Analysis

Deployment Level	No. of Workers	Cost per Unit (₹)	Total Cost (₹)
Prototype	1	2,100	2,100
Pilot Deployment	10	1,800	18,000
Zone-Level Deployment	100	1,400	1,40,000
City-Level Deployment	1,000	1,050	10,50,000

Cost reductions are achieved through economies of scale and system optimization.

Cost reduction is calculated using the following relation :
Reduction % = (2100 − cost per unit) ÷ 2100 × 100
So:

- Pilot (₹1,800) → (300 ÷ 2100) × 100 = **~14% reduction**
- Zone (₹1,400) → (700 ÷ 2100) × 100 = **~33% reduction**
- City (₹1,050) → (1050 ÷ 2100) × 100 = **exactly 50% reduction**

These reductions aren't random — they map to the bulk procurement savings we established in Section 4. At 1,000 units, components like the LoRa module drop ~50%, PCB integration saves another chunk, and assembly costs fall — all of which collectively bring the per-unit cost down by exactly half.

7. Operational and Maintenance Costs

Category	Estimated Cost
Device Enclosure (Rugged, Waterproof)	₹500 per device

Maintenance and Calibration	₹300 per device/year
Battery Replacement	₹200 per device/year
Miscellaneous Servicing	₹200 per device/year
Total Annual Maintenance Cost	**₹700–₹1,000 per device**

8. Total Cost of Ownership – 5-Year Projection (100 Units, Zone Level)

The table below consolidates all cost components — hardware, maintenance, cloud infrastructure, and contingency — into a single 5-year Total Cost of Ownership (TCO) view for zone-level deployment (100 workers).

Cost Item	Year 1 (₹)	Year 2 (₹)	Year 3 (₹)	Year 4 (₹)	Year 5 (₹)
Hardware (100 units @ ₹1,400)	1,40,000	–	–	–	–
Annual Maintenance (100 × ₹850)	85,000	85,000	85,000	85,000	85,000
Cloud Infrastructure	12,000	12,000	12,000	12,000	12,000
Contingency Reserve (10%)	23,700	9,700	9,700	9,700	9,700
Annual Total	**2,60,700**	**1,06,700**	**1,06,700**	**1,06,700**	**1,06,700**
Cumulative Total	**2,60,700**	**3,67,400**	**4,74,100**	**5,80,800**	**6,87,500**

Key observations:

- Year 1 cost is highest due to one-time hardware procurement of ₹1,40,000.
- From Year 2 onwards, annual expenditure drops to approximately ₹1,06,700.
- The 5-year cumulative TCO is approximately ₹6,87,500 for 100 workers — less than ₹1,375 per worker per year.
- A 10% contingency reserve is included to account for component price fluctuations, damaged units, and re-calibration requirements.

9. Software and Infrastructure Cost

Prototype Phase

- Firebase (Database and Hosting): Free
- Backend (Node.js): Open-source
- Dashboard Hosting: Free

Total Prototype Software Cost: ₹0

9.1 Production Phase (Firebase Blaze Plan)

Estimated usage for 100 workers:

- Data per message: ~200 bytes
- Messages per minute: ~30
- Monthly data usage: ~24 GB

Component	Monthly Cost (₹)
Database Usage	200–500
Cloud Functions	100–300
Hosting and Bandwidth	200–500
Total Monthly Cost	**₹500–₹1,500**
Per Worker Cost	**₹15–₹20/month**

10. Phased Deployment Timeline

The Veer Suraksha Grid is designed for phased rollout, allowing progressive learning, cost optimization, and risk mitigation at each stage.

Phase	Timeline	Scale	Key Activities	Est. Budget (₹)
Phase 1: Prototype & Pilot	Months 1–3	1–10 workers	Finalize device design, calibrate sensors, run pilot with select SMC sanitation workers. Identify field issues.	₹20,000–₹25,000
Phase 2: Zone-Level Deployment	Months 4–9	Up to 100 workers	Bulk procurement of 100 units. PCB integration. Deploy across one municipal zone. Activate dashboard and cloud monitoring.	₹1,40,000–₹1,60,000
Phase 3: City-Level Deployment	Months 10–18	Up to 1,000 workers	Full city rollout. Optimized cost through large-scale procurement. Comprehensive real-time governance dashboard operational.	₹10,00,000–₹11,00,000

This phased approach ensures that each stage is validated before scaling, reducing implementation risk and enabling iterative improvement.

11. Comparison with SMC Budget

Category	Allocation
Sanitation Worker Salaries	₹215+ crore
Machinery	₹1 crore
Sewer Infrastructure	₹128+ crore

Comparative Analysis

Parameter	Current System	Proposed System
Safety Monitoring	Minimal	Real-time
Worker Safety Investment	Negligible	₹1,000–₹2,000
Decision System	Manual	Data-driven
Monitoring	Fragmented	Centralized

Financial Insight:

- Less than 0.01% of the SMC budget is required for full deployment.
- ₹1 crore can equip more than 40,000 workers.

12. Cost-Benefit Analysis

Factor	Impact
Worker Safety	Significantly improved
Accident Reduction	High
Emergency Response	Faster
Governance	Transparent and accountable

The system cost is minimal compared to:

- Worker compensation claims
- Medical expenses
- Legal liabilities
- Operational inefficiencies

13. Return on Impact

Preventing a single incident can save:

- ₹5–10 lakh in compensation
- Medical and legal costs
- Productivity losses

Compared to a device cost of only ₹2,100 per worker (prototype) or ₹1,050–₹1,400 at deployment scale, the system delivers exceptional return on public investment.

14. Financial Feasibility

The proposed system is:

- Cost-effective and scalable
- Compatible with existing municipal infrastructure
- Suitable for phased deployment
- Sustainable for long-term operation

The 5-year TCO for a zone-level deployment of 100 workers is approximately ₹6,87,500 — an expenditure well within existing municipal allocations and negligible relative to the human cost of preventable accidents.

15. References

- Solapur Municipal Corporation Budget 2025–26
- Firebase Pricing Documentation – firebase.google.com/pricing
- Robu.in – Component retail pricing used to establish prototype BOM costs
- ElectronicsComp – Component pricing benchmarks for bulk reduction estimates
- Government of India – The Prohibition of Employment as Manual Scavengers and their Rehabilitation Act, 2013

16. Implementation & Sustainability Model (Minimal Margin Approach)

To ensure long-term feasibility, reliability, and sustainability of the Veer Suraksha Grid, a **minimal margin implementation model** is proposed for deployment under the Solapur Municipal Corporation (SMC).

16.1 Model Overview

The system is not positioned as a profit-driven commercial product. Instead, it follows a **cost-efficient, service-oriented model**, where the implementation team operates as a **technology and maintenance partner** to SMC.

Under this approach:

- The system is provided at **near-cost pricing**

- A **minimal margin** is included to ensure sustainability
- The focus remains on **public safety, reliability, and long-term operation**

16.2 Cost Structure

A. Device Cost (One-Time Deployment)

- Estimated bulk manufacturing cost: ₹950 – ₹1100 per device
- Proposed deployment cost (including minimal margin): ₹1100 – ₹1200 per device

The marginal difference is not intended as profit, but as a **sustainability buffer** to support operational continuity.

B. Annual Maintenance Cost (AMC)

A nominal annual service cost is proposed:

- ₹200 – ₹300 per device per year

This includes:

- Routine maintenance and calibration
- Battery servicing and minor replacements
- Device health monitoring
- Technical support and troubleshooting

16.3 Roles and Responsibilities

Solapur Municipal Corporation (SMC)

- Provides funding for deployment
- Owns the hardware infrastructure and system data
- Monitors system performance and compliance

Implementation Team (Technology Partner)

- Device manufacturing and supply
- System deployment and installation
- Dashboard and backend management
- Ongoing maintenance and technical support
- Periodic system upgrades and optimization

16.4 Justification for Minimal Margin

The inclusion of a minimal margin is essential for the following reasons:

1. **Operational Sustainability**
   Ensures availability of resources for repairs, replacements, and continuous system support
2. **Long-Term Reliability**
   Enables consistent maintenance, reducing system downtime and failures
3. **Scalability**
   Supports expansion to larger deployments (zone-level to city-level) without financial constraints

4. **Accountability**
   Establishes a responsible implementation partner committed to system performance

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16.5 Financial Impact

Even with the inclusion of a minimal margin:

- The overall system cost remains **significantly lower than conventional vendor solutions**
- The deployment cost remains **well within municipal budget limits**
- The cost per worker remains **economically viable for large-scale adoption**

17. Conclusion

The Veer Suraksha Grid presents a comprehensive, scalable, and economically viable solution for enhancing the safety and monitoring of sanitation workers under the Solapur Municipal Corporation (SMC).

Through detailed budget analysis, the system demonstrates that a prototype cost of approximately ₹2,100 per device can be reduced to nearly ₹950–₹1,100 at large-scale deployment through bulk procurement, communication module optimization, and integrated PCB design. This significant cost reduction makes the solution practical for city-wide implementation without imposing substantial financial burden on municipal resources.

The inclusion of operational cost analysis, cloud infrastructure estimation, and a 5-year Total Cost of Ownership (TCO) further establishes the system's long-term sustainability. Even at scale, the per-worker cost remains minimal when compared to existing municipal expenditures and the financial implications of workplace incidents.

The proposed minimal margin implementation model ensures that the system remains affordable while enabling continuous maintenance, accountability, and system reliability. By positioning the implementation team as a technology and service partner rather than a profit-driven vendor, the model aligns with public-sector priorities of cost efficiency, transparency, and long-term viability.

Overall, the Veer Suraksha Grid is not only a technically sound solution but also a financially feasible and socially impactful initiative. Its adoption has the potential to significantly improve worker safety, reduce risks, and strengthen governance through real-time monitoring and data-driven decision-making.

In conclusion, the system offers a practical pathway for transforming sanitation worker safety infrastructure in a cost-effective and sustainable manner, making it highly suitable for phased municipal deployment and future scalability.